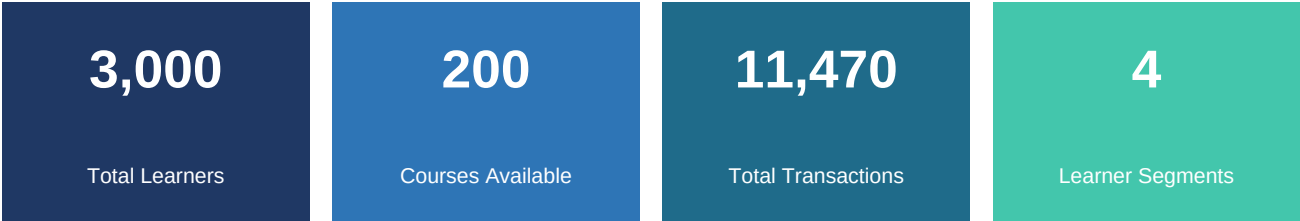


# EXECUTIVE SUMMARY

## Student Segmentation & Personalized Course Recommendation System

EduPro Online Learning Platform | Unified Mentor Internship | Jones Wesley | 2024



### Purpose & Business Context

EduPro serves a diverse global learner base but has historically relied on generic, undifferentiated course recommendations. This project builds a data-driven personalization engine that understands distinct learner types, recommends relevant courses at the right difficulty level, and supports targeted learner engagement strategies.

The system moves EduPro from a product-first to a learner-first platform architecture — a critical shift for improving course completion rates, reducing churn, and maximizing learner lifetime value.

### Analytical Approach

|                     |                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------|
| Data Sources        | Users (demographics), Courses (catalog metadata), Transactions (enrollment + spending history) |
| Feature Engineering | 10 behavioral features per learner: engagement, preference, and behavioral indicators          |
| Segmentation Model  | K-Means Clustering (k=4), validated with Agglomerative Hierarchical Clustering                 |
| Recommendation      | Cluster-aware content-based filtering with rating-weighted relevance scoring                   |
| Validation Metrics  | Silhouette Score, Inertia (Elbow Method), Engagement Lift per segment                          |
| Delivery Platform   | Interactive Streamlit dashboard with 5 modules for real-time exploration                       |

### Learner Segments — Key Findings

| SEGMENT 1   Casual Explorers (30.7% — 921 learners)  |                                                                                                  |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Avg Courses Enrolled                                 | 3.5                                                                                              |
| Avg Total Spend                                      | \$375                                                                                            |
| Preferred Category                                   | AI/ML                                                                                            |
| Preferred Level                                      | Beginner                                                                                         |
| Business Strategy                                    | Onboarding campaigns, gamification, free introductory courses to deepen commitment               |
| SEGMENT 2   Power Learners (22.5% — 675 learners)    |                                                                                                  |
| Avg Courses Enrolled                                 | 6.1 (highest of all segments)                                                                    |
| Avg Total Spend                                      | \$622 (highest of all segments)                                                                  |
| Preferred Category                                   | AI/ML                                                                                            |
| Preferred Level                                      | Advanced                                                                                         |
| Business Strategy                                    | Premium subscription bundles, certification tracks, cohort programs — highest ROI segment        |
| SEGMENT 3   Budget Browsers (21.3% — 640 learners)   |                                                                                                  |
| Avg Courses Enrolled                                 | 2.3 (lowest of all segments)                                                                     |
| Avg Total Spend                                      | \$156 (lowest of all segments)                                                                   |
| Preferred Category                                   | Cybersecurity                                                                                    |
| Preferred Level                                      | Advanced                                                                                         |
| Business Strategy                                    | Flash sales, discount vouchers, free preview content — high intent but price-sensitive           |
| SEGMENT 4   Focused Achievers (25.5% — 764 learners) |                                                                                                  |
| Avg Courses Enrolled                                 | 3.5                                                                                              |
| Avg Total Spend                                      | \$361                                                                                            |
| Learning Depth Index                                 | 1.45 (5x higher than Casual Explorers)                                                           |
| Preferred Level                                      | Advanced                                                                                         |
| Business Strategy                                    | Structured learning paths, skill badges, progressive course sequences — loyalty-building segment |

Model Performance Summary

| Metric                             | Value                    | Interpretation                                          |
|------------------------------------|--------------------------|---------------------------------------------------------|
| Silhouette Score (K-Means, k=4)    | 0.114                    | Meaningful cluster separation in high-dimensional space |
| K-Means vs Hierarchical Clustering | Consistent               | Both methods confirm 4-cluster structure                |
| Power Learner Engagement Lift      | +58% courses, +64% spend | Segment drives disproportionate platform revenue        |
| Budget Browser Engagement Lift     | -39% courses, -59% spend | Clearly distinguishable low-value segment               |

### Deliverables Completed

- analysis.py — Full EDA, feature engineering, clustering, and evaluation pipeline (9 steps)
- app.py — Streamlit dashboard with 5 interactive modules
- learner\_features\_clustered.csv — Engineered feature matrix with cluster assignments for all 3,000 users
- research\_paper.pdf — Detailed academic write-up with methodology, results, and references
- executive\_summary.pdf — This document
- requirements.txt — Reproducible dependency specification

### Strategic Recommendations for EduPro

- Prioritize Power Learners for premium product upsell — they represent the highest revenue potential per user
- Activate Budget Browsers with time-limited discount campaigns on Cybersecurity content — high intent, price-barrier only
- Build structured learning paths for Focused Achievers to increase retention and platform stickiness
- Deploy onboarding sequences for Casual Explorers with gamification to accelerate progression into higher-value segments
- Embed the recommendation engine as an API endpoint for real-time course suggestions on EduPro web and mobile apps
- Refresh cluster assignments quarterly as new transaction data accumulates

### Conclusion

This project delivers a production-ready intelligence layer for the EduPro platform. The combination of behavioral segmentation and personalized recommendation transforms EduPro's approach from reactive (showing all courses) to proactive (showing the right course to the right learner at the right time). The Streamlit dashboard makes these insights accessible to non-technical stakeholders without requiring data science expertise.

The framework is fully modular — new data can be reprocessed through `analysis.py` to refresh cluster assignments, and the recommendation engine automatically adapts to updated learner profiles. EduPro is now equipped with a scalable, interpretable, and actionable personalization foundation.